

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and implemented **AI agent orchestration** workflows using LangChain and a **vector database**, enabling dynamic task execution for enterprise risk assessments.
- Developed **Python** proof-of-concept applications to integrate emerging **LLM capabilities**, focusing on prompt engineering and function calling for risk mitigation scenarios.
- Engineered **production-ready AI capabilities** with **infrastructure-as-code** using **Crossplane** on **AWS**, ensuring repeatable deployments managed by platform teams.
- Configured and directed **AI coding assistants** like GitHub Copilot and Claude Code, integrating new internal AI tools and ensuring adherence to Marsh standards for software development.
- Acted as a subject-matter expert in **retrieval-augmented generation (RAG)**, coaching peers on implementation patterns and best practices for integrating external knowledge into LLM applications.
- Collaborated with Marsh business units through architecture forums to gather requirements for AI-driven solutions, translating needs into actionable proof-of-concept proposals.
- Containerized AI services using **Docker** and deployed them to a managed **Kubernetes** cluster on **AWS**, ensuring scalability and operational stability.

SKILLS

AI/ML – AI Engineering, Generative AI, LLM, Prompt Engineering, Retrieval-Augmented Generation (RAG), Agent Frameworks, Function Calling, Model Context Protocol (MCP), AI Safety, Guardrails, Fine-tuning, Model Adaptation, AI Coding Assistants

Programming Languages – Python, TypeScript, Rust, Go, Java

Cloud Platforms – AWS, Azure, GCP

Infrastructure & DevOps – Infrastructure-as-Code (IaC), Crossplane, Terraform, Pulumi, CDK, Kubernetes, Docker, CI/CD, Jenkins

Databases – Vector Databases, PostgreSQL, MySQL, Oracle

Software Engineering – REST APIs, Software Design, System Integration, Code Reviews, Agile Methodologies

PROJECTS

AI Agent Framework for Risk Analysis (HCL Technologies)

- Developed a proof-of-concept using **Python** and **LangChain** to build autonomous agents that analyze risk reports, integrating with a **vector database** for context retrieval.
- Implemented **prompt engineering** strategies to guide agent behavior, focusing on accuracy and adherence to predefined risk assessment protocols.
- Containerized the agent application using **Docker** for consistent deployment across development and staging environments.

RAG Implementation for Internal Knowledge Base (HCL Technologies)

- Built a **retrieval-augmented generation** system in **Python** that allows users to query an internal documentation repository using natural language.
- Integrated with **AWS Lambda** and **S3** to process and index documents, creating embeddings for efficient retrieval.
- Designed a simple REST API using **FastAPI** to serve query results, demonstrating a production-ready pattern for knowledge retrieval.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)